

NEUROSCIENCE

Human cortical networks trade communication efficiency for computational reliability

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Brains are often described as cost-efficient communication networks that optimally balance long-connection costs against fast communication. Inspired by the “use it or lose it” principle, we present a game-theoretic model of self-organizing neural units showing the brain is suboptimal in both regards. Regional competition for connectivity under propagative dynamics yields networks resembling the human cortex yet more efficient and economical. In addition, using a reservoir computing framework, we find comparable information processing capacity, but synthetic optimal communication networks show lower computational reliability. Last, virtual lesions reveal why these networks are fragile: To optimize communication, they funnel information through a spatially clustered “oligarchy” of transmodal hubs. The human brain instead uses a distributed “rich-club” backbone that better resists targeted attacks, despite higher wiring costs and less efficient communication. Cortical networks thus trade both cost and efficiency for reliable computation, highlighting computational reliability as an overlooked and perhaps even more prominent driver of brain connectivity than wiring cost or communication efficiency.

INTRODUCTION

What principles govern the intricate wiring diagram of the human brain? Aristotle and other natural philosophers, limited in experimental methods, established a top-down view that remained dominant for centuries: Form follows function. They believed the brain’s ventricles and gyrification pattern to have emerged to support its ultimate function, be it cooling the body or, as became increasingly apparent later, seating cognitive functions (1). However, over a century ago, Ramón y Cajal introduced a radically different idea by proposing a neural-level trade-off instead: Neural units in the brain must minimize costly connections while ensuring efficient communication (2). Networks dominated by short-range connections are cost-effective but suffer from limited communication capacity due to increased intermediate processing steps needed to move information through the network. Conversely, networks with abundant long-range connections benefit from enhanced signaling efficiency but with high metabolic and wiring costs (3). Thus, the human brain’s complex connectivity pattern, featuring numerous short-range connections and a few selective long-range shortcuts, is thought to represent the optimal balance of this trade-off (4–6).

Information flow within large-scale brain networks is constrained because not all regions are directly connected. Cognitive operations rely on efficient exchange of information between distributed neural populations, and cognitive difficulties can arise when that communication is disrupted (7–12). Yet, it is unclear what constitutes efficient

neural communication. The conventional approach rooted in graph theory assumes information to flow exclusively over the shortest path between two brain regions (13), here called routing communication dynamics. However, following recent advancements in mapping brain-wide activity patterns, a growing consensus surrounding alternative models of communication is forming. It is becoming clearer that brain regions do not interact this way because routing requires each region to have complete “knowledge” of the brain’s global network topology (14). Efforts to develop more decentralized models of interregional communication have broadly focused on propagation or diffusion dynamics rather than precise routing (7, 14, 15) (Fig. 1). In contrast to routing, propagation assumes information to cascade over all pathways while decaying with distance. Diffusion considers information as a random walker, serendipitously traversing the network until reaching its target. Thus, these alternative frameworks challenge traditional definitions of communication efficiency by assuming communication to unfold over multiple parallel (and longer) pathways instead of just the shortest one.

Under these models of neural signaling, could simple and organic processes lead to better network configurations, or is the human brain truly at the optimal balancing point of communication efficiency and wiring cost? In addition, does better communication naturally lead to better computation, or are there other distinct factors determining the brain’s connectivity pattern, e.g., computational capacity? We address these questions by combining game theory, communication models, generative brain network models, and neuromorphic computing. Game theory enables us to formally express the cost-benefit trade-off as a local and distributed optimization problem, where brain regions act as self-interested “players” negotiating connections without global network knowledge. We implement these interactions via a generative network model, where regions connect only if the benefit of improved signaling outweighs the wiring cost (Fig. 1). Notably, these decisions occur unilaterally as one region might find a connection advantageous whereas its target does not, resulting in asymmetric connectivity costs. Last, we use neuromorphic computing to test the computational properties of networks that result from different trade-offs.

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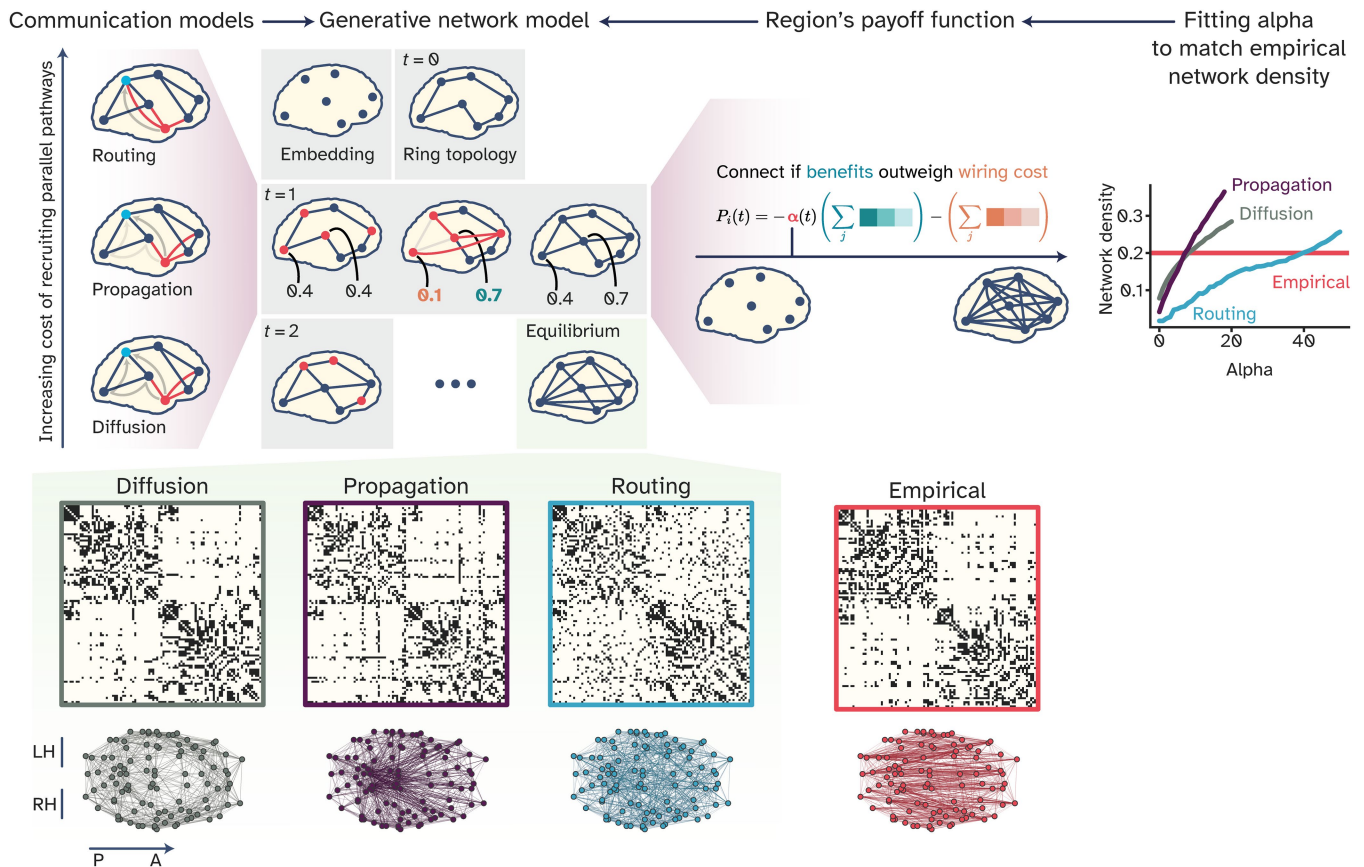


Fig. 1. Visual abstract of the framework. At each step of the model, a random group of nodes (red) evaluates its payoff function by comparing the current network configuration with an alternative where existing connections are removed and previously absent ones are added. Each node independently chooses whichever option yields a higher payoff, without considering effects on the rest of the network. The payoff function combines two components: wiring cost, measured as the Euclidean distance between connected nodes, and communication capacity, estimated through one of the three models of interareal communication. In the routing model, capacity depends only on the shortest path length between nodes; in propagation, it is based on broadcasting along all available paths, with longer routes contributing less; and in diffusion, it reflects the expected number of intermediate steps taken by random walkers. These models represent a spectrum from high to low parallel communication cost, with routing the most expensive, diffusion the least, and propagation in between. A trade-off parameter was then tuned to balance communication efficiency against wiring cost, calibrated to match the connection density observed in empirical brain networks. Note that the opacity of connections in the plotted brain networks is set proportionally to their length to emphasize long-range connectivity, which models disagree the most.

First, brain-like wiring patterns naturally emerged from competitive interactions among self-interested neural elements, each optimizing its own cost-benefit problem. Propagation-based networks produced structured long-range connectivity and even surpassed the human brain in both wiring economy and communication efficiency. Incorporating generated networks into a reservoir computing framework to solve a memory capacity (MC) task demonstrated that human cortical architecture offers superior computational capacity compared to both diffusion-based and routing-based but not propagation-based networks. Further analysis of propagation-based networks revealed that they optimally navigated the cost-benefit trade-off, primarily by establishing a transmodal “high-cost high-capacity backbone for communication” (16) akin to the brain’s rich-club organization. However, this backbone comprised fewer regions and was more spatially localized, resembling an “oligarchy” rather than a distributed rich-club connectivity. Consequently, we show that this centralization introduced both functional and signaling fragility that is averted by the human brain.

Collectively, these findings reveal that local competitive interactions naturally lead to the formation of complex network architectures,

some even surpassing the brain in both wiring economy and communication efficiency. However, the brain’s seemingly suboptimal wiring efficiency is offset by its exceptional computational reliability, suggesting an interplay between two evolutionary scales: top-down network-level pressures regulating bottom-up neural competitions.

RESULTS

To establish a normative benchmark for networks with optimal navigation of the trade-off between wiring cost and communication efficiency, we introduced a generative network model grounded in game theory. Briefly, our model begins with a spatially embedded set of regions minimally connected with one another, i.e., a ring graph (but also see figs. S2D and S3D). At each time point, a random subset of 10 regions is selected to “play” by disconnecting from their connected neighbors and connecting to random regions within the selected set. Each node then calculates and compares its payoff between the two configurations, opting for the more advantageous one (Fig. 1; also see section “Game theoretical generative network model”). Note that

regions do not consider their impact on other nodes nor do they care about the cost-efficiency of the network as a whole. This process continues until no new configuration is advantageous for any region and a state of equilibrium is reached (fig. S3).

The payoff function was comprised of two elements: the wiring cost of the established links, quantified as the sum of the Euclidean distance between node pairs, and the communication capacity, i.e., total nodal influence derived from three putative communication models. In routing, this capacity was inversely proportional to the shortest path length between nodes, assuming information flows solely through this pathway (7). Under diffusion dynamics, signaling is modeled as a random walker, navigating the network serendipitously until it reaches its destination. Here, communication capacity is based on the “effective distance” between pairs, defined as the expected number of intermediate nodes traversed by random walkers between a pair (17). Last, propagation assumes information to be broadcasted simultaneously along all pathways, with longer ones contributing proportionally less to communication capacity (18). Essentially, these models can be arranged along a spectrum of solutions to minimizing the energetic cost of parallel signaling with routing representing the most and diffusion the least expensive parallel communication (Fig. 1). A single trade-off parameter α was then adjusted to balance communication efficiency against wiring cost, ensuring that the networks have the same number of connections (density) as the empirical data (see Game-theoretic generative network model in Materials and Methods).

It is crucial to note that the goal of this framework is not to replicate the brain's connectivity pattern but to create networks where individual nodes achieve an optimal balance between wiring cost and signaling efficiency, conditioned on the underlying communication dynamics. Thus, the model remains fundamentally agnostic to the brain's connectivity pattern, relying solely on the spatial arrangement of nodes and the network's density. Therefore, the modeling framework may not necessarily converge to brain-like connectivity patterns but to a pattern that optimizes this trade-off. The resulting networks were then used as normative references and compared with empirical data to determine whether the brain also exhibits characteristics of an optimal communication network.

Empirical brain networks were constructed from 100 unrelated subjects provided by the Human Connectome Project (HCP) (19). These cortical networks were composed of 100 nodes defined on the basis of the Schaefer-100 parcellation atlas (20), with a network density of 20%. A group-averaged template of all empirical networks was created to benchmark empirical network similarity measures (Fig. 2, A and B, vertical band). Moreover, we constructed four null models: two through rewiring of empirical networks while keeping the degree distribution unchanged (randomized and latticized), one by rewiring the spatially defined long-range connections, and one fully random. These networks represent the extreme configurations of the cost-benefit trade-off. Randomized networks shuffle connectivity while only preserving the nodal distribution of connections, introducing many costly shortcuts. Latticized networks, in contrast, rewire topologically defined long-range connections, forming a web of mainly neighbor-to-neighbor connectivity, all while keeping the degree distribution unchanged. Although this results in networks with long processing pathways, they may still contain spatially defined long-range connections. We addressed this issue by rewiring spatially defined long-range connections (long-range rewired) and by constructing fully random networks with the same density as the

empirical network (Erdős-Rényi). Moreover, because the probability of observing long-range connections in the brain follows an exponential decay over distance (21), we explored this biologically plausible rule as another null model in fig. S5. To have the same sample size across settings, 100 synthetic networks per condition were generated. All networks were binary and undirected, and reported statistical relationships have $P < 0.01$ unless specified otherwise.

Self-organizing networks capture hallmarks of cortical architecture

We used a host of measures to compare generated networks with cortical and null networks, as shown in Fig. 2.

First, we asked whether connectivity in the brain shows characteristics of an efficient communication network. We used “endpoint similarity” defined as the cosine similarity between the connection pattern of the same node in two networks (e.g., emerged from routing-based dynamics versus the brain) to address this question. A value of one represents the highest similarity, i.e., when a region is linked exactly to all the same regions as in the target network, and no similarity results in zero (see Materials and Methods). Averaging across all nodes, we found that networks constrained by all three communication dynamics better capture the empirical data compared to randomized, latticized, and fully random networks (between 0.5 and 0.6 versus 0.2; Fig. 2A). This is also visible in fig. S3A for a comparison based on principal components analysis (PCA) embedding, which resulted in a similar ranking. Moreover, modeled networks also performed better than null networks in predicting individual links in the empirical brain, given two measures of accuracy (80% versus 65%) and F1 score (between 0.5 and 0.6 versus 0.2; Fig. 2B). Accuracy measures the overall percentage of correct predictions (true positives and negatives), whereas F1 scores take possible class imbalance into account by taking the harmonic mean of precision and recall instead. Both metrics show moderate but considerably better alignment, compared to other null networks. Note that the null network generated by rewiring long-range connections is closest in all three metrics, which is expected given that the rest of the network is identical to the empirical data. However, null networks based on the exponential decay rule are as close to the brain as the synthetic networks (fig. S5), which is due to the replication of the same “decay over distance” trend in these models (Fig. 2C). Last, the highest achievable endpoint similarity and F1 score, measured by comparing the empirical data with a subject-averaged template is about 0.8 and 90% for accuracy. Together, direct comparison of the cortical wiring patterns with those simulated on the basis of connectivity-agnostic self-organizing models showed that the brain contains signatures of an efficient communication network.

To then test which topological features of empirical networks are captured by our generated networks, we measured small-worldness, average clustering coefficient, structural complexity, wiring cost, and average path length. Briefly, average clustering coefficient is a measure of segregation or “cliquishness” in the network. Small-worldness measures the balance of this segregation and global integration in a network. Values around zero show a perfect balance, with negative values representing more clustered and segregated networks and positive values hinting at greater integration resulting from many shortcuts. Average path length measures the expected number of intermediate nodes in processing pathways and is the conventional measure of signaling efficiency under routing dynamics. Structural complexity measures the spread of the network's eigenspectrum. Larger

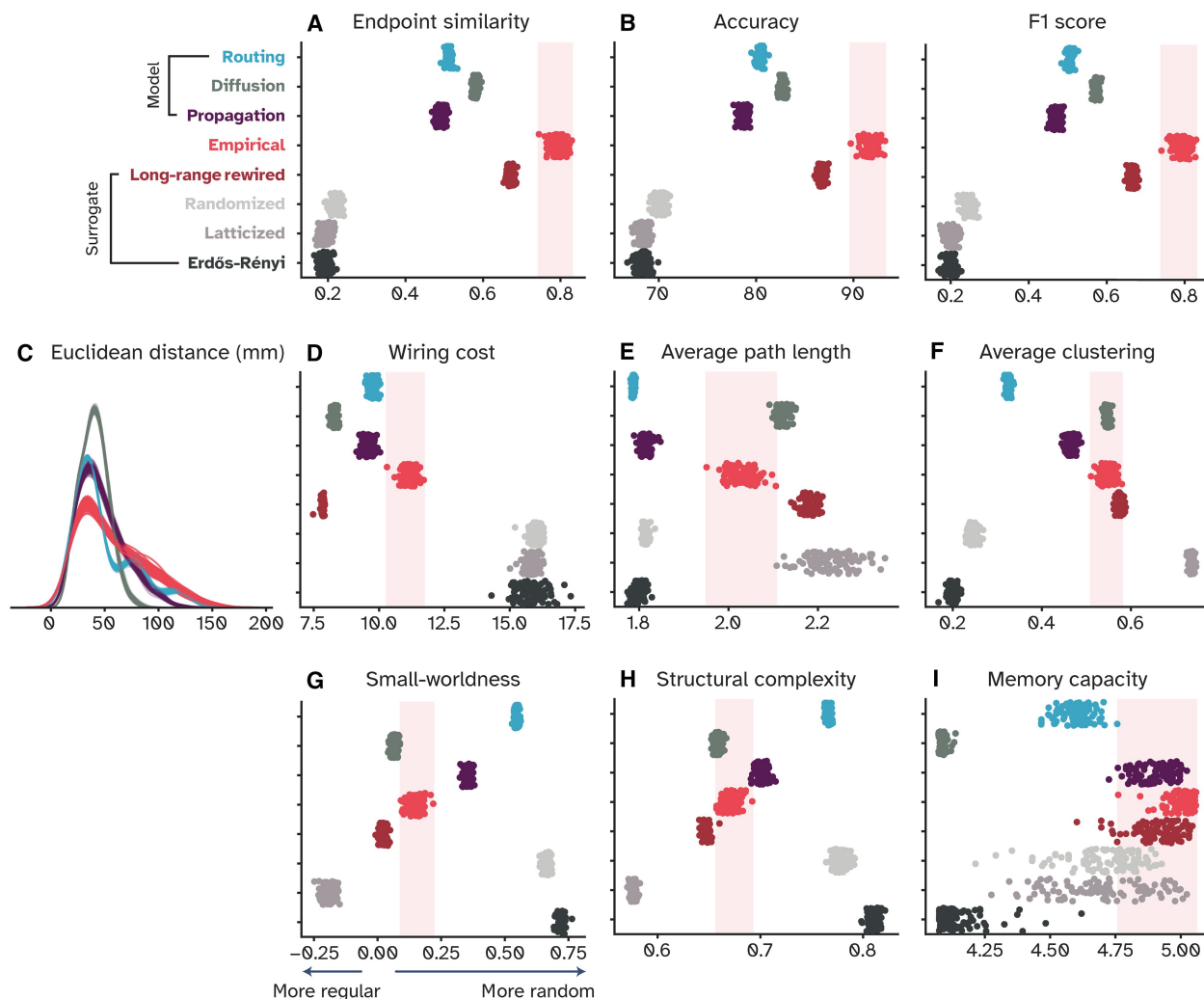


Fig. 2. Optimal communication networks capture hallmarks of brain connectivity. Each point represents one network, and the empirical range is shown by the shaded red area. (A and B) Model networks capture aspects of empirical cortical wiring evident from the fact that endpoint similarity, accuracy, and F1 scores show better alignment with empirical data than randomized, latticized, or Erdős-Rényi nulls. Individual empirical networks are compared with the group-averaged consensus connectome (C) distribution of Euclidean connection lengths across network types. Diffusion-based networks lack long-range connectivity, whereas routing results in precise wiring that leads to three peaks. Propagation leads to similar long-tail decay but with more short-range and fewer long-range connections compared to empirical data. (D to F) Comparison of wiring cost, average path length, and clustering coefficient shows that the human brain is not optimal in either wiring economy or communication efficiency, as traditionally defined on the basis of average path length. (G and H) Small-worldness and structural complexity highlight differences in global organization, with diffusion-based models resembling empirical topology despite lacking long-range connectivity. (I) MC results show that empirical networks outperform most generative and null models.

complexity corresponds to a more diverse distribution of eigenvalues, hinting at a larger pool of possible neural dynamics (22, 23). Together, these plots (Fig. 2, E to H) can be interpreted as follows: Networks constrained by diffusion dynamics exhibit brain-like topologies as they tend to align more closely with empirical values across nearly all measures. For instance, average clustering coefficient of both spans from 0.5 to 0.6. However, these model networks lack long-range connections, suggesting that the high resemblance is driven by the abundance of short-range connections (Figs. 1 and 2C). Although all three modes of communication result in more cost-efficient wiring patterns compared to the brain, only diffusion results in an average shortest path length that exceeds that of the brain (Fig. 2, D and E). This indicates that, according to traditional

measures of graph efficiency, diffusion leads to lower communication capacity than the brain due to its lack of long-range shortcuts, although it precisely mirrors the brain's average clustering coefficient. We found that null networks based on exponential decay rule follow the same trend (fig. S5), suggesting that networks based on diffusion dynamics might be following this rule more closely than the others. Notably, both routing and propagation achieved efficiency levels of random networks while still maintaining more parsimonious wiring cost (Fig. 2, E and D).

Last, networks derived from routing dynamics tend to exhibit far more randomness than those of the brain (Figs. 1 and 2, F and G), with clustering and small-worldness values aligning more closely with random graphs rather than the brain. Collectively, these findings

indicate that, first, the brain, although still cost-efficient, is not optimally managing either its wiring cost or its communication capacity as better configurations were achieved. Second, constraining diffusion dynamics results in networks lacking long-range connectivity, yet it captures other topological characteristics of the brain better. In contrast, routing produces networks that appear random and fail to replicate the brain's notable features, such as its average clustering coefficient and small-worldness. Last, propagation yields more efficient networks, achieving better wiring economy, comparable structural complexity, clustering, and small-worldness to that of the brain.

Each of our simulated networks was optimizing a different communication dynamic, but what was the impact of this optimization on that network's computational properties? We evaluated all networks using a neuromorphic computing paradigm known as echo state networks (ESNs). These models resemble conventional artificial neural networks, particularly recurrent neural networks, but differ primarily in their training approach (24). In ESNs, the hidden layer, or “reservoir,” remains unaltered, whereas only the readout layer is tuned to solve the task (see Materials and Methods, Reservoir computing framework). To estimate the computational capacity of our networks, we used the widely recognized MC task (25–28), where the network receives a sequence of random values as input and must recall lagged versions of it. The further back in the sequence it can reproduce, the better the MC. Previous research with brain-inspired reservoirs has yielded mixed results. Although some found brain-inspired networks to outperform random connectivity, others found the opposite (27–30). It is therefore unclear whether more efficient communication, as conventionally measured on the basis of routing dynamics and attributed to random connectivity, leads to enhanced computational capacity or not. As we have networks specifically optimized for communication, we used them to address this open question. If more efficient communication naturally translates to better task performance, then we should expect MC to follow the rough ranking of average path length (Fig. 2E), with networks characterized by shorter average path length performing better. However, MC exhibited a different ranking, with the human brain at the top, followed by propagation-based models and the long-range rewired null model (Fig. 2I). Notably, diffusion-based networks consistently underperformed compared to other networks, including null models. In addition, fully random networks performed poorly compared to randomized and latticized brain networks (see Fig. 2I, the bottom three categories), suggesting that the brain's heterogeneous degree distribution, which is also present in these null models, plays a role in computation as well.

We replicated these results via two approaches: one with a different parcellation scheme based on the Desikan-Killiany atlas with a total of 68 cortical regions (fig. S1) and the other with 100 subjects from another dataset (HCP young adults; fig. S5). In the latter, we also included 32 subcortical regions derived from the Tian atlas (31) to ensure that the results remain valid despite introducing subcortical networks (fig. S7).

In summary, this section highlights two key findings: First, under propagative signaling dynamics, selfish nodes that optimize their local cost-communication trade-off naturally form networks that exhibit characteristics of human cortical connectivity and computational capacity. In addition, these models surpass the human brain in terms of wiring economy and communication efficiency, suggesting a suboptimal connectivity pattern in empirical data. Together, propagation-based networks are more efficient and cost-effective than the brain

while also maintaining comparable computational capacity, establishing a relatively better wiring configuration. This begs a question: Could the “suboptimal” cortical connectivity of the human brain be attributed to other selective pressures than wiring economy, communication efficiency, and computational capacity?

Human cortical networks prioritize robustness over communication efficiency and wiring economy

Emergence of structured long-range connectivity, cost-effective wiring, comparably high MC, combined with results from previous works (32–34) permit us to consolidate propagation as the plausible regime of communication and focus further on the relationship between propagation-based networks and the empirical ones. Doing so, we plotted MC as a function of global “propagation efficiency” (PE) (but see fig. S2A for routing and diffusion) and found that the human brain resides at the Pareto front of these two objectives (Fig. 3A and fig. S12).

On the basis of Pareto optimality theory in evolutionary biology, organisms balance multiple and often conflicting goals, resulting in phenotypes that are not optimal for any single goal (35, 36). Instead, Pareto-optimal solutions are a range of solutions that dominate suboptimal ones and lie on the “Pareto front” of the solution space, each prioritizing different objectives of the trade-off (37, 38). In our case, an optimal solution unconstrained by a trade-off between MC and PE would occupy the top-right corner of the plot (indicated by the yellow circle). However, all network topologies are confined to a space with Pareto front solutions lying on an arch, suggesting that, after a point, marginal increase in PE leads to a rapid decline in MC (see fig. S2A for a clearer visualization). Pareto rank analysis found that following a fully random network, the human brain, as well as propagation and routing-based networks occupy the largest surface of the arch-shaped front (fig. S12). Notably, this arch contains an unstable inflection point between two stable zones of “high performance, low communication efficiency” and “low performance, high communication efficiency.” On the left-hand side of this zone lies the human brain with reliably high performance, whereas on the right-hand side, performance tends to decay as PE increases. Said differently, networks resulting from both propagation and routing communication regimes sit on the unstable inflection point, whereas the human brain networks are spread relatively flat across the high-performance zone, prioritizing reliable computation across multiple instances, i.e., different subjects [see fig. S12 (left) for a clearer visualization]. This suggests that computational reliability may be a factor driving human cortical architectures away from better communication efficiency because further increasing communication pushes networks toward the unstable tipping point.

Collectively, Fig. 3A suggests that the human brain could further optimize signaling efficiency and still remain at the Pareto front as propagation-based networks do. However, that neighborhood of the Pareto front appears to be an unstable tipping point, beyond which marginal gains in PE leads to a sharp decline in MC. By prioritizing reliable computation, human cortical networks sacrificed both more economical wiring configurations and enhanced communication. This finding aligns with previous research showing that spatially embedded networks perform better and show characteristics of cortical networks when PE is minimized as opposed to maximized (39, 40).

Next, we aimed to uncover potential reasons behind the functional instability of the propagation-based networks, i.e., why individual differences in instantiation of the same generative mechanisms

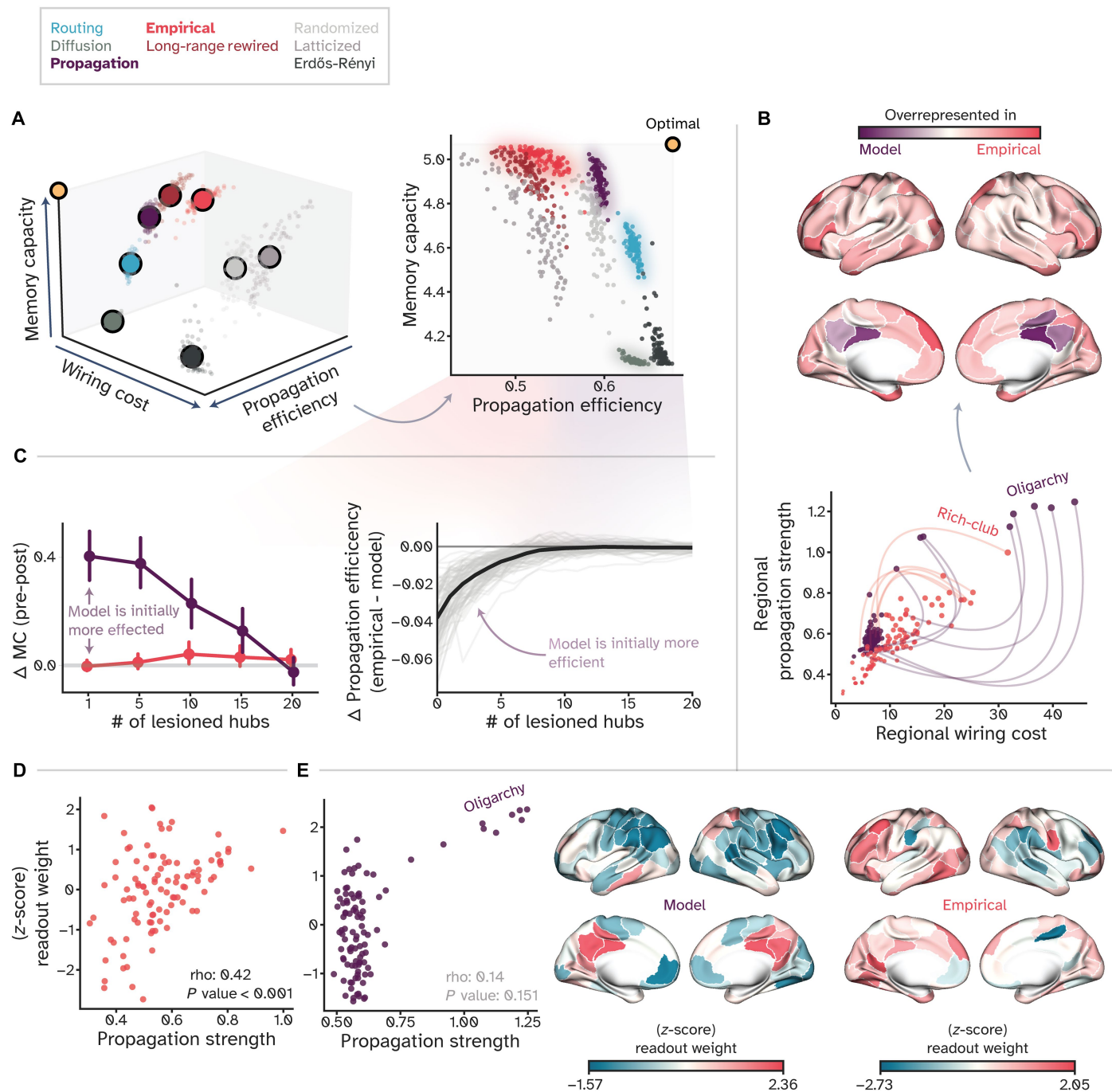


Fig. 3. Distributed hub connectivity makes brain networks less efficient but more robust against targeted attacks. (A) Relationships among wiring cost, communication efficiency, and MC reveal a trade-off between communication and computation, with the empirical brain prioritizing robust computation over efficient communication. (B) Comparison of regional cost-capacity values shows that the human brain allocates connections in a more distributed manner, forming a larger set of hubs (rich-club), whereas propagation-based models converge on a smaller and spatially confined set of oligarchs. Curved lines show discrepancy between high-degree nodes of the two networks, indicating their disagreement in allocating connections. (C) Progressive lesioning of hubs in these two networks shows how oligarchy forms a functional bottleneck to enhance information propagation. Empirical networks are less susceptible to performance degradation, whereas removing even one hub from the oligarchy reduces MC. (D) In reservoir networks based on empirical connectomes, nodes with stronger propagation strength also show higher functional relevance, measured via their allocated readout weight, whereas (E) in propagation-based networks, this relationship is absent. The functional importance is concentrated within the small oligarchic core, revealing the bottleneck again.

can lead to wide variation in MC? Comparing how these networks allocated resources relative to the brain (Fig. 3B) revealed a critical difference: spatially distributed rich-club organization versus a localized oligarchy.

Rich-club connectivity as a design solution against fragile efficiency

Rich-club organization is a hallmark of brain connectivity across multiple species (41, 42). This organization is formed when well-connected hub regions disproportionately connect to each other, establishing a “high-cost, high-capacity backbone of communication” (16). It is proposed that this backbone integrates information across different modalities (16, 43), enhances dynamic complexity of the network (44), and ultimately supports complex cognition (45, 46). Comparing empirical cortical networks with those based on propagation revealed a disagreement in allocating resources, i.e., connections, in the network (Fig. 3B, indicated by the purple and red lines). Although both networks have a spectrum of nodes from low-cost, low-capacity to high-cost and high-capacity regions, they appear to be differently distributed in propagation-based models (Fig. 3B and fig. S2B). Most regions in these models belong to the low-cost, low-capacity category, with a handful of extremely resourceful and influential regions forming a localized oligarchy instead of a spatially distributed rich-club (fig. S2B). We verified this pattern of connectivity does not depend on propagation strength (figs. S8 and S9), which is a parameter of the propagation communication model representing the ease of spreading information through the network. Smaller values of this parameter represent more abrasive network and larger values model smoother communication, i.e., higher propagation strength (see Materials and Methods for details). Figure 3B shows that the oligarch regions are clustered around the posterior cingulate cortex and precuneus, which were previously found to be core members of rich-club organization (43). However, in this case, they dominate connectivity (even when subcortical regions are included; fig. S7), prompting us to ask what is the computational role of the oligarchy? After all, this pattern of connectivity supports the same level of MC as the empirical brain.

Figure 3 (C to E) answers this question. We used a progressive lesioning scheme of the top 20 regions ranked by degree, tracking memory, and communication capacities of the network (Fig. 3C). Although the human brain remains functionally robust throughout the lesioning, perturbing the oligarch regions resulted in a drastic reduction of both communication efficiency and MC. The perturbation effect became comparable when the oligarch regions were fully removed, suggesting a restabilization of the network after the bottleneck is gone (Fig. 3C and fig. S6). This type of response to perturbation, known as the “Sprague effect” (47, 48), was previously observed in both human brains and neural networks when disruptions caused by a lesion are alleviated following subsequent lesions to other nodes (49).

To provide a mechanism for why perturbing oligarch regions has a disproportionate effect on computation, we focused on the assigned readout weights to each region and compared them against their propagation strength (Fig. 3, D and E) because these weights are the only trainable component of our reservoir computing framework. We hypothesized that the network’s hub regions have better representations of the input signal, given their extensive connectivity and capacity to integrate information from a wider range of regions than peripheral nodes. This hypothesis was inspired by the finding that long-range connectivity, specifically among rich-clubs,

enhances dynamic complexity of the brain (44, 50–52). We found a significant correlation between nodal propagation capacity and readout weight (Fig. 2D) in cortical networks ($\rho = 0.42$) but not in propagation-based networks ($\rho = 0.14$, P value = 0.15), meaning that the readout layer was predominantly relying on the oligarchy (Fig. 3E). Moreover, to verify that the dynamic complexity of the network is related to this finding, we devised a series of subsequent analyses summarized in figs. S10 and S11 in which we directly tracked the dynamic complexity of the network. We found that there is an inverted U-shaped relationship between communication efficiency and computational capacity. We also found that propagation-based networks are easier to synchronize, further suggesting that network synchronizability may be a potential mechanism for this trade-off. Our results support previous findings and the overall conclusion that hyperefficient networks show lower information processing capabilities because information spreads through them quickly without having the opportunity of being properly processed (44, 53, 54). Collectively, it becomes clear that oligarchy is the main driver of the relatively high computational capacity seen in propagation-based networks. Consequently, lesioning them initially degrades task performance because the overall repertoire diversity of the network is reduced (fig. S11).

Next, to pinpoint how this wiring pattern emerged from local competition for influence, we performed a cost-benefit analysis for each connection, asking whether the connection justifies its cost for both sides it connects. Briefly, for each edge, we computed the payoff of both nodes, removed the connection, and compared the payoffs with and without the connection. If removing the connection makes one side better off and the other worse, it is likely that the link was established by the region benefiting from it (i.e., the one that is worse off when the link is removed). The results reveal that nodes in the oligarchy would actually profit from link removal, indicating that this centralized core had emerged due to decisions made by peripheral nodes (Fig. 4A and fig. S2C). More notably, the spatially proximate neighbors of the oligarchy benefit the most from this arrangement (Fig. 4A). In other words, regions closest to the oligarchy pay very little wiring cost connecting to them while gaining easy access to the whole network via the oligarchy’s extensive connectivity. We then noticed that the topographical organization resulting from this analysis appears to broadly resemble a smooth gradient separating unimodal from transmodal regions (Fig. 4B). Thus, we asked whether these peripheral regions are the sensorimotor unimodal areas (55). Because previous research has shown a disproportionate evolutionary and developmental uptake in wiring of transmodal areas (56–59), answering this question could provide further evidence for local and communication-based connectivity optimization to be a key contributor.

Therefore, we compared the cortical map of regional cost-benefit values with well-established natural axes of cortical organization: the sensorimotor-association (SA) axis (56), the unimodal to transmodal functional hierarchy (60), the first principal axis of cortical activation maps during 123 cognitive tasks (Neurosynth) (61, 62), and the cross-species functional homology between humans and macaque monkeys (63). Significant correlations in all four comparisons supported the notion that the oligarchy is formed by human-distinct regions of association cortices (Fig. 4B). In essence, the hyperconnected oligarch-club appears to have emerged to enhance communication among, and integrate information from, unimodal regions. However, in the model, and unlike the brain’s rich-club, the circle was further restricted to a smaller set of apex regions to optimize wiring and communication even more. Notably, we also found modest correlations with maps of cortical

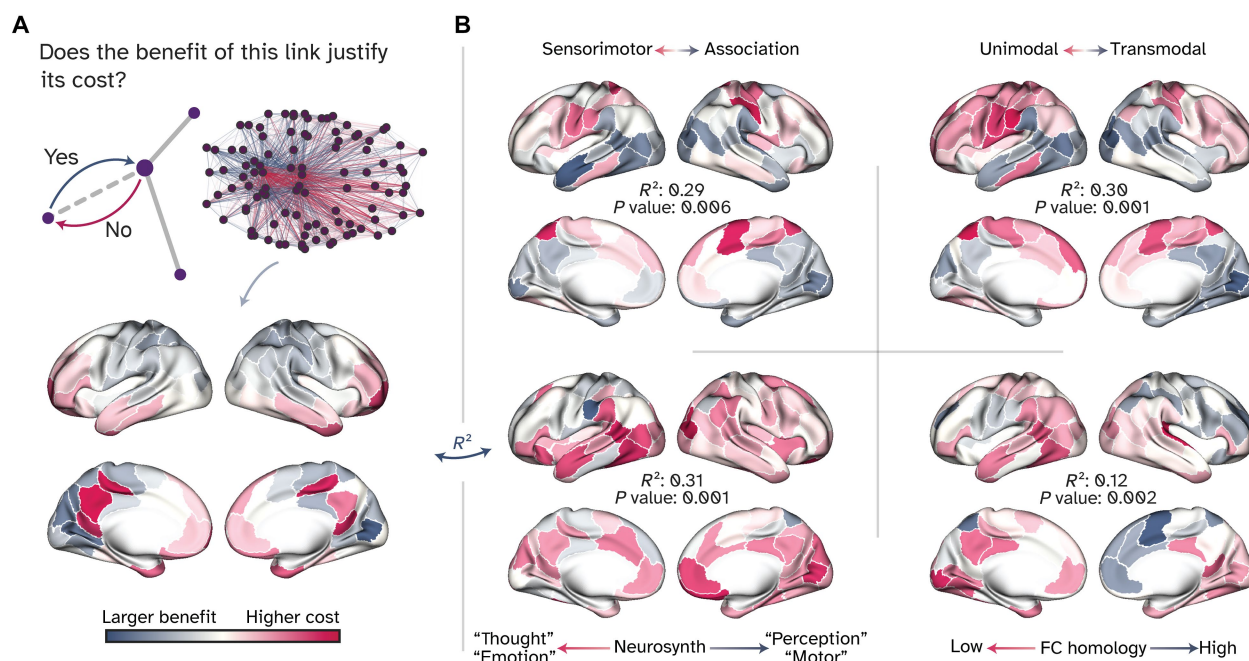


Fig. 4. Cost-benefit analysis of individual links and its network-wide organization. (A) Cost-benefit analysis of individual connections reveals that oligarchic hubs emerge primarily from peripheral nodes' aspiration to integrate with the rest of the network without paying the wiring cost. Cortical projection of this cost-benefit analysis highlights that the oligarchy is spatially clustered around posterior midline regions, e.g., not only the precuneus and posterior cingulate cortex but also the temporal poles and lateral orbitofrontal cortices. (B) These regional differences align with established cortical gradients, including the unimodal-transmodal, SA, large-scale maps of cortical activation during various cognitive tasks (table S1), and cross-species functional homology axes. Reported P values are adjusted for spatial autocorrelation (see Materials and Methods, Null maps for statistical significance).

thickness and gene expression ($R^2 = 0.16$ and 0.27 , respectively, P value = 0.02) while finding no relationship with maps of cortical scaling, mean regional blood flow, the second axis of functional gradient, and evolutionary cortical expansion (fig. S4). Together, significant correlations with maps related to functional arealization of the brain (SA) alongside nonsignificant correlations with cortical expansion and scaling suggest the following: Although the connectivity pattern of transmodal regions was, in part, shaped to integrate incoming information from distant unimodal areas, cortical scaling and expansion were driven to meet the required processing demands.

To summarize, results from this section provide an answer for why propagation-based networks are relatively computationally powerful but less resilient: Optimizing for communication and wiring economy resulted in architectures that naturally develop integrator hub regions, enhancing their capacity for diversifying representations. However, this organization is also a tight bottleneck, sometimes leading to diminished MC as seen in the previous section (Fig. 3, A, C, and D). Moreover, given that oligarch regions are among the brain's rich-club regions, it could be that, among other factors, external top-down pressure for robustness constrained the enrichment of those regions further into the oligarchy. In other words, the oligarchy appears to be an extreme case of current rich-club organization that can emerge from unrestricted bottom-up local competition for connectivity (Fig. 5).

DISCUSSION

We revisited a well-established hypothesis suggesting that a trade-off between communication efficiency and wiring cost is the primary driving force constraining the brain's complex connectivity

pattern. Using a game-theoretic generative network framework, constrained by putative communication models, we found that patterns resembling cortical connectivity emerge only under propagative signaling dynamics. Optimal communication networks under routing dynamics place precise long-range connections, resulting in far more random wiring patterns than the brain, whereas diffusion-based networks contained no long-range connectivity (Fig. 2C). This wiring pattern was previously reported when optimizing for wiring cost by replacing regional positions instead, with the same detrimental effect on communication efficiency (5, 6). Together, our findings are consistent with previous research, which indicates that interareal communication in brain networks follows a broadcasting-like dynamic, rather than the shortest path routing or random walker-based diffusion (32, 33).

In our modeling framework, individual regions negotiate their own local trade-offs by competing for influence, making the global topology fully emergent. That is, we only fit the model for the number of connections, not for topological characteristics as is commonly done (37, 64, 65). This local competition approach was inspired by the well-studied phenomenon of "use it or lose it," wherein neural elements compete for resources and can eventually even take over each other's functions (66, 67). Our results suggest that this type of competition also supports the emergence of complex patterns such as those seen in the human cortex. Notably, this simple model, constrained by propagation-like signaling dynamics, generated networks that have a better wiring economy than the brain. This design solution was achieved by forming a centralized oligarch-club instead of a distributed rich-club, leading to lower wiring costs, more efficient neural communication, all while maintaining a computational capacity on

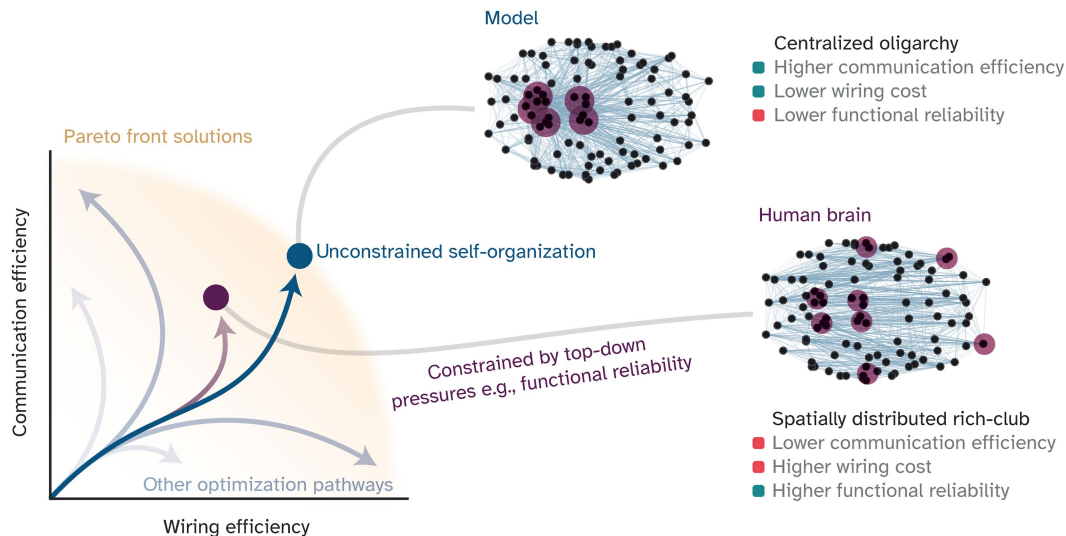


Fig. 5. Visual summary of the main findings. Although unconstrained local competition for signaling efficiency drives networks toward a hyperefficient but fragile “oligarchy” (blue), the human brain (purple) settles for suboptimal configurations that maintain better functional resilience. By adopting a spatially distributed “rich-club” architecture, cortical networks sacrifice better wiring economy and communication efficiency to ensure robust computation, revealing resilience as a critical top-down constraint on brain organization. This schematic is based on the PCA embedding of optimization trajectories shown in fig. S3.

par with the empirical brain networks. However, we show that the hyperconnected oligarchy is also an information chokepoint. Consequently, these networks concentrate readout weights on just the oligarchy, show more variability in task performance across different instantiations, and degrade faster with synthetic lesions.

Together, these findings add reliable computational capacity as another component to the cost-efficiency trade-off, where the human brain invests resources and sacrifices communication efficiency to consistently achieve a high information processing capacity. A recent work on large-scale recording from the prefrontal cortex and olfactory bulb in developing mice revealed an oligarchic organization where a small set of hub neurons show disproportionate activity (68). Notably, the study finds that this architecture is preconfigured rather than experience-driven. Together with their findings, our results suggest a key role for neural computation underlying behavior in distributing hubs across the cortex. Further research could model nonbehaving model organisms such as brain organoids (69) to test whether an oligarchic organization emerges naturally and sustains when there is no external pressure on neural units to coordinate for supporting behavior.

It is important to clarify what our model was not intended to do: On the basis of game theory, this framework is considered normative, i.e., provides a possible answer for why the human brain is organized as such and not how (70, 71). The difference is that our framework is not equipped to model neurobiological mechanisms by which neural units are assumed to navigate their trade-offs. Our results indicate that maximizing communication could be a selective pressure on neural units, constrained by wiring cost and the dominant regime by which information flows. On the basis of previous findings (64, 72–76), we hypothesize homophily, i.e., higher preference for connectivity among similar neural units, to be a candidate mechanism implementing the local goal of achieving maximal communication. However, maximizing homophily instead of communication as a local optimization objective result in networks

with extreme core-periphery architectures where all nodes link only to their local hubs and not each other (fig. S3C). It is also important to keep in mind that, whereas the decision-making process, i.e., navigating the trade-off, takes place at the local level instead of the whole brain, the algorithms used to compute them are not local. For instance, in our framework, for a node to compute its payoff under all models of dynamics, it still is informed about how much information it transmitted to all other units (see Materials and Methods). Together, although our framework answers a teleological question, more work is needed to find biologically plausible mechanisms through which this trade-off could be implemented. Last, our model is not an evolutionary algorithm nor are the results necessarily generalizable across different species. It is, however, of great interest to use datasets from other species to see whether, for instance, organization of nonvertebrate nervous systems better aligns with an optimal communication network. One could further investigate how the trade-off between communication efficiency and wiring cost is navigated by other brains than humans.

For simplicity and mathematical tractability, our work was limited to binary and symmetric networks. Future work could address both limitations by introducing weights and asymmetric relationships. We expect that introducing an asymmetric wiring cost where no region can impose costs on others to result in a faster convergence (77, 78). Here, we specifically opted out of asymmetric interactions because our networks of interest, i.e., empirical brain data, were also symmetric. Further research using connectomes derived from invasive tract tracing of other animal species could fully embrace the asymmetric version of our framework (79). However, introducing weights requires a shift from discrete to continuous games (80). These analytical complications might have novel machine learning-based solutions. For instance, graph neural networks could be used to directly predict the resulting network without the need for simulating the game. Another technical challenge that benefits from further investigations involves generating more realistic null networks

to better understand which features map to which trade-offs. Here, we used an established family of null networks although future works could expand it by considering that most interhemispheric connections pass through the corpus callosum. Last, it is also useful to note that, although our model is based on game theory, the equilibrium state is not a unique Nash equilibrium but a stable field around a point in which macroscopic statistical properties are tightly clustered while their microstructure varies (fig. S3D). This is particularly relevant for modeling individual differences because all brains are different despite the macroscopic properties of the human brain being largely similar. Further research is needed to see whether microscopic differences in the model are like those seen across different individuals and, if yes, what do they tell us about trade-offs influencing individual differences.

We primarily measured MC, assuming it serves as a foundational computation that provides a backbone for many other downstream functions. Future work could measure other tasks such as working memory (81) or navigation (82) to see whether the expanded trade-off landscape still shows the negative impact of hyper efficient communication. Nonetheless, these tasks are all computations performed by neural network models, and although using these models as measures of computational capacity is gaining attraction in neuroscience, they complement long-standing and more established approaches. One such approach, known as the critical brain hypothesis, posits that the brain operates near a phase transition (critical point) between order, when activity dies out quickly, and disorder, when activity explodes in the network (83). Criticality optimizes the information processing capacity of the network, providing a strong incentive for why the brain should position itself near this critical point (84). Similarly, human brain balances functional segregation and integration, which also maximizes information processing capacities by diversifying neural dynamics (44). Together, these two frameworks provide a complementary definition of computation grounded in network dynamics. On the one hand side of the spectrum lies subcritical and segregated networks, and on the other side sits supercritical and hyperintegrated networks. In both extremes, neural dynamics are not maximally complex and useful for downstream computations. This complexity peaks at an intermediate range (criticality) where there is also a balance between functional segregation and integration, leading to a maximal diversity in patterns of generated neural activity (fig. S11). Not only do we replicated our results using this definition (figs. S10 and S11) but also, through systematic rewiring of a toy network from a hyperefficient configuration to an inefficient one, we established that this relationship is not unique to the human brain. Our results, showing that propagation-based networks are more synchronizable than the human brain, align well with previous demonstrations of information spreading in complex networks. In these studies, hyperefficiency of communication leads to rapid synchronization of distant nodes (44, 85) and degrades the diversity of information because nodes parrot each other instead of encoding a rich set of features (86, 87).

Last, our lesioning experiment suggested a link between a network's topology and paradoxical lesion effects. Although connectivity has been a central focus of this topic (88, 89), how hubness influences these effects is not yet clear. Future research could shine light on this relationship by systematically perturbing ESNs with different hub organization.

To summarize our key findings (Fig. 5), local and competitive interactions among neural units can lead to the emergence of complex global connectivity patterns, capturing hallmarks of cortical

architecture. Moreover, reliable functional capacity is another organizing feature of the brain, leading to a less efficient and costlier rich-club organization as opposed to an oligarchy.

MATERIALS AND METHODS

Neuroimaging dataset and its preprocessing pipeline

For the main analysis, we used data from the HCP (90), restricted to the 100 unrelated subjects from the S900 release, and for our replication dataset, we used 100 separate subjects from the HCP young adults dataset (fig. S5). All participants provided informed consent in line with the HCP consortium protocols. More information about the imaging protocols and preprocessing pipeline is available in (91) and (92).

Diffusion-weighted MRI acquisition

The diffusion-weighted acquisition protocol is covered in (93). Briefly, diffusion magnetic resonance imaging (MRI) scans were collected on a Siemens 3T Skyra scanner using a spin-echo EPI sequence with a multiband factor of 3. The acquisition parameters were voxel size = 1.25 mm isotropic, TR = 5500 ms, and TE = 89.5 ms. Three diffusion shells were acquired with b values of 1000, 2000, and 3000 s/mm², each with 90 diffusion directions, plus six b_0 images.

Preprocessing and fiber reconstruction

Minimally preprocessed data from the HCP pipeline (93) were used, which included correction for head motion, eddy currents, and susceptibility artifacts. Diffusion-weighted data were reconstructed using q -space diffeomorphic reconstruction (QSDR) (94) in DSI Studio, yielding spin distribution functions in MNI space with a 1-mm resolution. Deterministic streamline tractography was then performed with a modified FACT algorithm (95) using the following parameters: angular cutoff = 55°, step size = 1.0 mm, minimum streamline length = 10 mm, maximum length = 400 mm, and anisotropy threshold set by Otsu's method. For each subject, 1,000,000 streamlines were reconstructed.

Parcellation atlases and network construction

Structural connectivity was estimated using one primary and another replication (fig. S1) cortical atlases, as well as another containing subcortical regions (fig. S5). For the primary atlas we used the Schaefer-100 parcellation that defines regional borders based on functional gradients and clustering of resting-state functional MRI (fMRI) connectivity (20). For the replication atlas (fig. S2), we used the anatomically defined Desikan-Killiany atlas, comprising 68 cortical regions (96). Streamlines were assigned to regions according to their termination points. For the replication dataset including subcortical regions, we used the combination of Schaefer-100 and Tian-S2 parcellation, resulting in 132 regions altogether.

For each subject and parcellation, a structural connectivity matrix was constructed by counting the number of streamlines between each pair of regions. Last, matrices were binarized after proportional strength-based thresholding to achieve a target density of 20%, consistent with prior studies (34, 92). Specifically, we used an algorithm provided by the Python library *netneurotools* that trims weak connections while preserving a backbone of minimum-spanning tree to ensure that the network remains connected. Together, this pipeline produced subject-specific connectomes for 100 subject that formed the empirical basis for comparison with null models and generative networks.

Game-theoretic generative network model

We implemented a game-theoretic generative framework in which each node i of the network acts as a self-interested player, locally

optimizing a payoff function that maximizes communication capacity given the wiring costs. The network is built first as an embedded ring graph where each node is connected to its two immediate neighbors. The embedding is based on the three-dimensional (3D) coordinates corresponding to the centroid of each region. At each iteration, a randomly selected set of 10 nodes evaluate their payoffs under the current configuration and compares it with the payoff of an alternative configuration where their connections are rewired. Rewiring simply flips the state of the given connection, meaning that, within the selected set, it cuts the existing connection and link unconnected nodes. If the alternative yields higher payoff, the node unilaterally updates its connections, i.e., keeps the new configuration. If not, the node defers the new link and keeps the current configuration (Algorithm 1). Because nodes decide unilaterally, new configurations can be imposed on those who did not choose it. Implications of this possibility were investigated through our cost-benefit analysis. This procedure is repeated for $T = 10,000$ steps, although an equilibrium state is usually reached within the first 3000. This equilibrium is visible in fig. S3A where the “developmental trajectory” of each model is plotted. It is also noticeable in fig. S2D where we show three random initial networks with different densities eventually settle on the same target density, i.e., the equilibrium. The payoff U_i for node i is defined as

$$U_i = \alpha \sum_j d_{ij}^{\text{comm}} + \sum_j c_{ij}^{\text{wire}}$$

where d_{ij}^{comm} is the communication distance between nodes i and j , and c_{ij}^{wire} is the Euclidean distance or the wiring cost of the edge. Communication distance is computed across the whole network because the objective is to be as influential as possible. The parameter α is the trade-off parameter controlling the relative weighting of communication over wiring. Effectively, α controls the density of the emerged network and is fine-tuned using a bisection search with linear interpolation to match the density of the empirical networks as 0.2 ± 0.01 (fig. S3C). The resulting α values are shown in Table 1.

Differences in α values can be interpreted as how much long-range connections should be discounted by their added value in efficiency term to be considered. For instance, under a routing dynamic, very long connections are needed to provide enough influence for a node to establish the connections; otherwise, not having more connections is preferred, which leads to an equilibrium at a sparser network. Last, we chose to formalize communication capacity as a cost (see below) so it can be jointly minimized with wiring cost.

Communication models

We considered three putative models of interregional communication: **Routing (formalized as the shortest-path distance)**

Information is assumed to travel exclusively along the shortest path between nodes i and j using Dijkstra’s algorithm

$$d_{ij}^{\text{routing}} = \min_{\pi \in \mathcal{P}_{ij}} \sum_{(u,v) \in \pi} w_{uv}$$

where $\mathcal{P}_{ij}$ is the set of all paths connecting i and j , and w_{uv} is the edge weight between nodes u and v . In this work, each edge has unit length, and the shortest path reduces to the minimal number of hops.

Diffusion (formalized as the effective distance)

Signaling is modeled as a random walk, where effective distance is proportional to the commute time between two nodes, i.e., the time

Algorithm 1. Game-theoretic generative network model.

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1: Inputs:
2:   Node coordinates,  $\mathbf{X} \in \mathbb{R}^{N \times 3}$ 
3:   Initial adjacency,  $\mathbf{A}^{(0)} = (a_{ij}) \in \{0, 1\}^{N \times N}$ 
4:   Number of iterations,  $t_{\max}$ 
5:   Trade-off parameter,  $\alpha$ 
6:   Communication metric,  $d^{\text{comm}} \in \{d_{\text{routing}}, d_{\text{diffusion}}, d_{\text{propagation}}\}$ 
7: Subroutine PAYOFF( $i, \mathbf{A}$ ):
8:   function PAYOFF( $i, \mathbf{A}$ )
9:      $C_{\text{wire}} \leftarrow \sum_{j \in \mathcal{A}_i} \|\mathbf{x}_i - \mathbf{x}_j\|_2$ 
10:     $C_{\text{comm}} \leftarrow \sum_j d_{ij}^{\text{comm}}(\mathbf{A})$ 
11:    return  $-\alpha C_{\text{comm}} - C_{\text{wire}}$ 
12: end function
13: Subroutine UPDATE( $\mathbf{A}, S$ ):
14:   function UPDATE( $\mathbf{A}, S$ )
15:      $\mathbf{A}' \leftarrow \mathbf{A}$  ▷ work on a copy of  $\mathbf{A}$ 
16:     for all  $i, j \in S$  do ▷ Remove edges within the selected subset  $S$ 
17:       if  $A'_{ij} = 1$  then
18:          $A'_{ij} \leftarrow 0$ 
19:       if  $A'_{ji} = 1$  then ▷ if graph is undirected
20:          $A'_{ji} \leftarrow 0$ 
21:       end if
22:     end for ▷ Add / rewire edges from each  $i \in S$  to random nodes within  $S$ 
23:     for all  $i \in S$  do
24:       choose a target node  $k \in S \setminus \{i\}$  uniformly at random
25:        $A'_{ik} \leftarrow 1$ 
26:        $A'_{ki} \leftarrow 1$  ▷ if graph is undirected
27:     end for
28:     return  $\mathbf{A}'$ 
29: Main routine:
30: for  $t = 1$  to  $t_{\max}$  do
31:   Sample a subset  $S \subseteq V$  of size  $|S| = 10$  uniformly at random
32:    $\mathbf{A}' \leftarrow \text{UPDATE}(\mathbf{A}^{(t-1)}, S)$ 
33:   Compute payoff difference:
34:    $\Delta P \leftarrow \sum (\text{PAYOFF}(i, \mathbf{A}') - \text{PAYOFF}(i, \mathbf{A}^{(t-1)}))$ 
35:   if  $\Delta P \geq 0$  then
36:      $\mathbf{A}^{(t)} \leftarrow \mathbf{A}'$ 
37:   else
38:      $\mathbf{A}^{(t)} \leftarrow \mathbf{A}^{(t-1)}$ 
39:   end if
40: end for
Output: Emergent adjacency  $\mathbf{A}^{(t_{\max})}$  and network metrics

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Table 1. Optimal trade-off parameter of different models.

	Schaefer-100	Desikan-Killiany
Routing	43.01	39.25
Diffusion	8.13	4.90
Propagation	10.46	9.79

it takes for a walker to go from one node to the other and returns (17). Using the graph Laplacian L , the resistance distance is defined as

$$d_{ij}^{\text{diffusion}} = (\mathbf{e}_i - \mathbf{e}_j)^T L^+ (\mathbf{e}_i - \mathbf{e}_j)$$

where $\mathbf{e}_i$ and $\mathbf{e}_j$ denote the standard basis vectors corresponding to nodes i and j , respectively, defined as column vectors with a single entry of 1 at the designated index and 0 elsewhere. In this context,

the difference $(e_i - e_j)$ acts as a current operator injecting unit flow at node i and extracting it at node j . L^+ is the Moore-Penrose pseudo-inverse of L .

Propagation (formalized as Katz distance)

Information is assumed to broadcast simultaneously over all possible paths, with contributions from longer paths monotonically discounted (97). The resulting propagation matrix P is computed from

$$P = (p_{ij}) = (I - \beta A)^{-1} (I - \beta A)^{-T}$$

where A is the unweighted adjacency matrix and β is a scaling factor, here set to 0.7, which matches the empirical brain propagation strength (32). However, we found that these results are robust within a wide range of β values (from 0.4 to 0.95; see figs. S8 and S9). The communication distance is then

$$d_{ij}^{\text{propagation}} = -\log P_{ij}$$

Homophily (formalized as topological distance)

As an additional topological constraint, we quantified homophily, defined as the similarity between the connection profiles of nodes (64, 73, 74) and computed using the matching index of adjacency rows

$$d_{ij}^{\text{homophily}} = 1 - \frac{A_i \cdot A_j}{|A_i| |A_j|}$$

where A_i is the binary connectivity vector of node i . Homophily reflects the tendency of nodes with similar nodal neighborhoods to connect preferentially.

Null networks

To benchmark our generative networks, we constructed three null models that modify the empirical brain connectome systematically (98). These models are randomized, latticized, and long-range rewired. In addition, we constructed fully random graphs with the same rough network density ($p=0.2$) as the empirical brain (fig. S3B) using the Erdős-Rényi algorithm.

Randomized

Topologically randomized networks were generated by rewiring the original connectivity matrices using the Maslov-Sneppen randomization algorithm (99, 100). This procedure preserves both the network density and the degree distribution while otherwise randomizing its topology. Concretely, pairs of edges (i, j) and (u, v) were selected and swapped such that the new connections became (i, v) and (u, j) . To ensure sufficient randomization, the rewiring step was repeated 250 times for each connection. Repeating for each connectome, the result is an ensemble of 100 randomized networks that retain certain local connectivity constraints, i.e., the number of connections per region, but lack the global organization of empirical networks.

Latticized

Topologically latticized networks were generated using a complementary procedure that biases the Maslov-Sneppen rewiring algorithm toward a ringlike topology while maintaining degree distribution and graph density (100). Specifically, swaps of edge pairs were only accepted if the resulting connections placed nonzero entries closer to the main diagonal of the empirical connectivity matrix (which often, but not always, corresponds to connections between spatially adjacent cortical regions). By iteratively favoring such swaps (250 times per connection), the algorithm yields networks in which each

node is primarily connected to its nearest topological neighbors, approximating a ring-lattice connectivity pattern. This produces highly ordered connectivity patterns with low global efficiency and high clustering (Fig. 2, E and F), providing a structure in contrast with the randomized condition. As with the randomized networks, this process was applied to all 100 empirical connectomes, yielding 100 latticized version.

Long-range rewired

In addition to randomized and latticized null models, we implemented a rewiring procedure designed to reduce the wiring cost of empirical networks by selectively swapping long-range connections to isolate their influence on different graph properties depicted in Fig. 2. Starting from the empirical adjacency matrices, we identified all edges whose length (measured via the Euclidean distance) exceeded 70 mm (Fig. 2C). These were classified as “long-range” connections. These edges were then sorted by distance in descending order, such that the longest connections were preferentially targeted. At each rewiring step ($N=10$), a subset of the remaining long-range edges was removed and replaced with a shorter alternative. Specifically, for a given edge (i, j) exceeding the distance threshold, the connection was deleted, and each node was evaluated for potential reconnection to its geometrically nearest unconnected neighbor. Whichever node (i or j) had the closer candidate neighbor was selected, and a new edge was formed. Applied to all empirical connectomes yielded 100 networks with no long-range connectivity. However, other connections were left identical to the corresponding empirical networks.

Density-matched exponential decay rule

To isolate the contribution of distance-dependent wiring from other topological features of the empirical connectomes, we generated a set of synthetic networks in which the probability of connection between any two regions decayed exponentially with their Euclidean distance. For each subject, the connection probability between regions i and j was defined as $\exp[-d(i, j)/\lambda]$, where $d(i, j)$ denotes the Euclidean distance between region centroids and λ is a length constant set to the median interregional distance across all node pairs. Edges were then sampled, without replacement and in order of decreasing probability, until the network reached a target density of 0.2, matching other connectomes. This procedure was applied independently to all 100 subjects, yielding 100 networks.

Erdős-Rényi

Last, we generated a set of 100 Erdős-Rényi random graphs in which each pair of nodes was randomly connected with independent probability $p=0.2$, yielding a binomial degree distribution and no higher-order topological constraints. This ensemble of ER graphs serves as a pure random baseline because, although randomization and lattice procedures preserve specific properties of the empirical data (degree distribution or spatial embedding), the ER model establishes expectations for network topology when both constraints, and payoff dynamics are absent.

Reservoir computing framework

Echo state networks

To assess the computational capacity of networks, we implemented them as the core of ESNs (24, 101). In this setup, the given networks served as the recurrent weight matrices defining reservoir connectivity. The reservoir state was updated according to the equation

$$x(t+1) = f[W_{\text{res}}x(t) + W_{\text{in}}u(t)]$$

where $x(t)$ denotes the reservoir state vector at time t , $u(t)$ is the external input, W_{res} is the spectrally normalized reservoir connectivity derived from the underlying network, and f is a nonlinear activation function.

For all experiments, we used the hyperbolic tangent ($\tanh$) as the nonlinear activation function. This choice ensures that the reservoir state trajectories are bounded while still providing rich nonlinear dynamics. The resulting reservoir states, harvested in response to the temporal input sequences, are later used to train a linear readout that performs the memory task described in the next section.

To enable fair comparison across networks, key parameters were held constant. The spectral radius of W_{res} was fixed at 0.99, ensuring near-critical dynamics, whereas input weights W_{in} were drawn from a uniform distribution in $[-1, 1]$ and scaled by a factor of 10^{-5} . Spectral normalization of W_{res} is essential to ensure that the dynamics remain stable and in the echo state regime: Without it, the system may become unstable (if the spectral radius exceeds 1) or lose expressivity (if it is too small). This normalization allows the reservoir to retain memory of past inputs while remaining sensitive to new inputs. The scaling of input kept the perturbations small relative to the internal reservoir dynamics, maintaining consistency across different network instantiations. Each network was tested with an identical set of random input sequences (see below). For robustness, we report results as the average performance across 256 independent trials for each network.

MC task and the training paradigm

We quantified reservoir performance using the well-established MC task (25–28, 101, 102). At each trial, the network received a 1D signal $u(t)$, constructed as an i.i.d. sequence of length 3000 drawn from $\mathcal{N}(0, 1)$. The network was trained to reproduce delayed copies of the input, with distinct output units assigned to different delay values τ where $\tau = \{1, 2, \dots, 40\}$.

For all lags τ , the readout mechanism, i.e., a linear regression model, was fitted to predict the delayed input $u(t - \tau)$ from the current reservoir state $x(t)$. The fitting solution was done analytically using the pseudoinverse of the reservoir state matrix

$$W_{\text{out}} = YX^+$$

where X is the matrix of reservoir states, Y is the target, where each Y corresponds to a specific delay τ , and X^+ denotes the Moore-Penrose pseudoinverse. Before training, the first 100 transient steps were discarded to wash out initial conditions. No regularization was applied to any of the networks.

A train-test split of 0.8/0.2 was used, with training performed on the first 80% of time points and performance evaluated on the remaining 20%. MC at each lag was quantified as the squared correlation coefficient between the predicted and target sequences. Last, the total MC of a network was obtained by summing ρ across all tested lags

$$\text{MC} = \sum_{\tau=1}^{40} \rho_{\tau}$$

with larger MC values indicating better task performance.

Network measures

Structural complexity

Prior research has shown that large-scale neural signaling is well described as a network of coupled linear systems (32, 103, 104). This

allows us to analyze the magnitudes of eigenvalues λ_i derived from networks' adjacency matrices to determine the stability and asymptotic regional decay in response to input. A broad distribution of $|\lambda_i|$ hints at the existence of multiple timescales (105, 106), indicating heterogeneous neural response profiles. Thus, to summarize the network's capacity for such rich temporal responses, we computed the entropy of its eigenspectrum. Because all our networks were symmetric, the eigenspectrum contains only real parts thus

$$p_i = \frac{|\lambda_i|}{\sum_{j=1}^n |\lambda_j|}$$

$$H = - \sum_{i=1}^n p_i \log p_i$$

Last, to normalize values between 0 and 1, we divided by the maximum possible entropy

$$\text{Structural complexity} = \frac{H}{\log n}$$

Average clustering coefficient

To assess the local segregation of networks, we measured the average clustering coefficient as follows. For each node i with degree $k_i \geq 2$, the local clustering coefficient C_i is defined as the fraction of existing edges among its neighbors relative to the maximum possible number of such edges

$$C_i = \frac{2T_i}{k_i(k_i - 1)}$$

where T_i denotes the number of triangles (three-node closed loops) including node i , and $k_i(k_i - 1)/2$ is the total number of possible neighbor-neighbor connections. For nodes with $k_i < 2$, C_i is set to zero. The average clustering coefficient of the network is then

$$C = \frac{1}{N} \sum_{i=1}^N C_i$$

where N is the number of nodes. C quantifies the tendency of nodes to form tightly interconnected neighborhoods. High C indicates the presence of dense local circuits or modules, whereas low C implies a more globally integrated but locally sparse topology. In brain networks, elevated clustering is typically interpreted as a signature of localized processing and functional segregation.

Average path length

To measure global integration, we computed the average path length of each network using the same shortest-path distances described above in subsection "communication models" under "routing." For any pair of nodes (i, j) , let $d(i, j)$ denote the length of the shortest path between them. The network's average path length is then simply

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d(i, j)$$

where N is the number of nodes. L measures the expected number of steps required to connect two arbitrary nodes in the network. Smaller L indicates more efficient global integration under the routing

signaling dynamics as information can traverse the network in fewer hops.

Small-worldness

To quantify small-world organization, we used the Telesford ω index (107), which positions each network on a continuum between lattice and random topologies. For a given network with clustering coefficient C and characteristic path length L , and two sets of degree-constrained ensembles of 100 null networks (described above): (i) a randomized ensemble yielding L_{rand} and (ii) a latticized ensemble yielding C_{latt} .

Together, ω is defined as

$$\omega = \frac{L_{\text{rand}}}{L} - \frac{C}{C_{\text{latt}}}$$

By construction, $\omega \in [-1, 1]$, where values near 0 indicate small-world networks (high clustering like a lattice but short paths like a random graph), $\omega \rightarrow -1$ indicates lattice-like organization with very high C and long L , and $\omega \rightarrow +1$ indicates random-like organization with low C and short L . Both these cases are visible in Fig. 2G.

Wiring cost

To quantify the cost of physical embedding for a pattern of connectivity, we computed the wiring cost as the Euclidean length of each edge in 3D space. Node positions were derived from the centroid coordinates of each cortical region in the parcellation. For any edge connecting nodes i and j with coordinates $r_i = (x_i, y_i, z_i)$ and $r_j = (x_j, y_j, z_j)$, the wiring cost is

$$d(i, j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

The average wiring cost of the network is the sum of edge lengths across all connections

$$W = \sum_{(i,j) \in E} d(i, j)$$

where E is the set of edges, and the average wiring cost is obtained by dividing W by $|E|$, the total number of edges.

Endpoint similarity

To assess how closely the connectivity patterns of corresponding nodes matched across two networks, we used endpoint similarity. For a given node i , we represent its connectivity profile as a binary adjacency vector a_i (row of the adjacency matrix). Endpoint similarity between networks A and B for node i is then defined as the cosine similarity

$$S_i = \frac{a_i^{(A)} \cdot a_i^{(B)}}{|a_i^{(A)}| \cdot |a_i^{(B)}|}$$

where $\cdot$ denotes the dot product and $|\cdot|$ is the Euclidean norm. Values range from 0 (no overlap in connectivity patterns) to 1 (identical connectivity). The network-level endpoint similarity is obtained by averaging across all nodes

$$S = \frac{1}{N} \sum_{i=1}^N S_i$$

where N is the number of nodes in the network. Endpoint similarity quantifies the extent to which networks match in their node-wise connectivity profiles, providing a direct comparison of regional wiring patterns between empirical, null, and generative networks.

Accuracy and F1 score

To evaluate how well modelled networks predicted the presence or absence of empirical connections, we used two conventional classification metrics: accuracy and F1 score. Given the set of all possible node pairs, predictions fall into four categories: true positives (TP: correctly predicted edges), true negatives (TN: correctly predicted non-edges), false positives (FP: predicted edges that do not exist), and false negatives (FN: missing predictions of existing edges). Accuracy measures the overall proportion of correctly classified pairs

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

F1 score balances sensitivity and precision by computing the harmonic mean of precision and recall

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Briefly, accuracy is influenced by the overall class balance (i.e., number of edges versus non-edges), making it useful for a global measure of prediction quality. In contrast, F1 score is more sensitive to class imbalance as it emphasizes the trade-off between precision (avoiding false positives) and recall (avoiding false negatives). This makes F1 informative when the number of absent connections greatly exceeds the number of present ones, as is the case in our sparse networks.

Graph distance

We used DeltaCon distance (108) to further measure the effect of different propagation strengths (Katz β parameter) on the structure of the generated networks (fig. S9). DeltaCon quantifies the dissimilarity between two graphs defined on the same node set by comparing their node-to-node affinity matrices derived from Fast Belief Propagation. For a graph with adjacency matrix A and degree matrix D , the affinity matrix is given by $S = (I + \epsilon^2 D - \epsilon A)^{-1}$, where $\epsilon = 1 / [1 + \max(D)]$ controls the attenuation of influence with topological distance, such that nearby nodes contribute more strongly to one another's affinity than distant ones. The distance between two graphs G_1 and G_2 with affinity matrices S_1 and S_2 is then computed as the Matusita (root-Euclidean) distance

$$d(G_1, G_2) = \sqrt{\sum_{i,j} (\sqrt{S_{1,ij}} - \sqrt{S_{2,ij}})^2}$$

which satisfies the axioms of a metric and is sensitive to changes in connectivity that propagate beyond direct edges. We computed DeltaCon distances between each generated network and its corresponding empirical connectome across the full range of β values to verify that our results were not contingent on a specific choice of propagation strength.

Dynamic complexity

In addition to MC, we measured dynamic complexity of the networks that provides a more general definition of neural computation,

specifically applicable to large-scale neural dynamics (44). Dynamic complexity, originally introduced as functional complexity (44), indexes the balance between integration and segregation by comparing the empirical distribution of pairwise correlations among regional activities to a uniform reference distribution. Concretely, the off-diagonal entries of the pairwise correlation matrix are binned into B equally spaced intervals over $[-1, 1]$, and the deviation of the resulting histogram from uniformity is computed as

$$C(A) = 1 - \frac{B}{B-1} \sum_{k=1}^B |p(r_k) - \frac{1}{B}|$$

where $p(r_k)$ is the relative frequency of correlations falling in bin k and the prefactor $B/(B-1)$ normalizes the measure to the unit interval. Networks whose activity is either fully synchronized or fully desynchronized yield correlation distributions concentrated in a narrow band, hence low complexity, whereas networks supporting a broad mixture of weakly and strongly correlated regional pairs yield near-uniform distributions and a correspondingly high complexity (fig. S10). For each network, dynamic complexity was computed from activity patterns generated using a linearized Wilson-Cowan model described in (32). Briefly, regional activity $x_i(t)$ evolved according to the multivariate Ornstein-Uhlenbeck process

$$\tau \frac{dx_i}{dt} = -x_i(t) + g \sum_{j=1}^N A_{ij} x_j(t) + \sigma_{in}$$

where A_{ij} is the connectivity between regions j and i , τ sets the intrinsic relaxation time of each node, g scales the global coupling strength, and ξ is the independent Gaussian noise with zero mean and unit variance scaled by σ_{in} . The system was integrated using the Euler method with time step $\Delta t = 0.001$ s for a total duration of 60 s, using $\tau = 0.01$ s, $g = 1.0$, and $\sigma_{in} = 1.0$. The resulting time series were used to compute the pairwise correlation matrix from which dynamic complexity was derived as defined above.

Synchronizability

Synchronizability quantifies how readily a network of identical oscillators can settle into a globally synchronized state, providing a structural proxy for the dynamical coordination that the topology can support. We computed synchronizability as the eigenratio of the combinatorial graph Laplacian

$$R = \frac{\lambda_N}{\lambda_2}$$

where λ_2 is the algebraic connectivity (Fiedler value, the smallest nonzero eigenvalue) and λ_N is the largest eigenvalue of $L = D - A$, with A the adjacency matrix and D the corresponding degree matrix. Although conventionally smaller values of R indicate a wider stable synchronization regime (an easier-to-synchronize network), here, to be more intuitive, we reported $1/R$ in fig. S10. Thus, in this plot, larger synchronizability values indicated more readiness for synchronization. For each network, L was constructed, its full eigenspectrum was obtained by symmetric diagonalization, and R was computed as λ_N / λ_2 .

Branching ratio and repertoire diversity

A growing body of evidence suggests that cortical circuits operate near a critical regime poised between subcritical, quickly extinguishing dynamics and supercritical, runaway activation. Networks tuned to this critical point are thought to maximize information processing

capacity by jointly supporting long-range propagation of activity, sensitivity to weak inputs, and a broad repertoire of distinguishable activity patterns. To probe how optimal propagation networks and the empirical human brain networks positioned their dynamics relative to this regime (fig. S11), we simulated a discrete three-state cellular automaton (the quiescent-excited-refractory, or QER, model) on these networks and quantified two complementary signatures: the branching ratio (BR) of activity cascades and the diversity of the spatial patterns those cascades produced. Following the formulation of Haimovici and colleagues (109), each region i occupied one of the three states $s_i \in \{0, 1, -1\}$ corresponding to QER [also known as susceptible-excited-refractory (SER)] (110), respectively. At every discrete time step, a quiescent node transitioned to the excited state if its summed input from currently excited neighbors

$$I_i(t) = \sum_j W_{ij} \mathbb{1}[s_j(t)=1]$$

exceeded a threshold T , or, with a small spontaneous activation probability r_1 , irrespective of its input. Excited nodes deterministically entered the refractory state at the next step and refractory nodes returned to quiescence with probability r_2 . We swept the threshold over $T \in \{10^{-3}, \dots, 10^1\}$ on a logarithmic grid of 100 values to span subcritical, critical, and supercritical regimes (fig. S11). We set $r_1 = 10^{-3}$ and $r_2 = 0.1$ in line with (109) and, at each T , averaged outcomes over 10 independent trials of 5000 time steps each, discarding the first 20 steps as transient.

The BR was computed as the per-step ratio of active nodes at consecutive time points

$$BR = \left\langle \frac{N_{t+1}^{ext}}{N_t^{ext}} \right\rangle_{t: N_t^{ext} > 0}$$

with values near unity identifying the critical point, $BR < 1$ indicating subcritical decay, and $BR > 1$ supercritical amplification. To quantify the variety of activity patterns each network could express, we extracted neuronal avalanches as contiguous time windows of nonzero global activity and represented every avalanche by the binary spatial pattern of regions recruited at any point during it (111). Duplicate patterns were removed, and for the resulting set of unique patterns, we computed pairwise normalized Hamming distances between all pairs and defined repertoire diversity as the median of this distance distribution. Higher repertoire diversity therefore indicates that the network visits a broader, less stereotyped set of coactivation patterns over the course of its dynamics (111).

Pareto rank analysis

Pareto rank quantifies how close each network is to being nondominated, expressed as an integer count of “layers” of strictly better networks ahead of it in objective space. A rank of 0 means the network sits on the frontier itself, whereas higher ranks indicate progressively greater distance from it. Formally, network i dominates network j in the joint maximization of MC and PE if $MC_i \geq MC_j$ and $PE_i \geq PE_j$ with at least one strict inequality. The frontier F_0 is then the set of nondominated networks across the pooled set ($n = 800$); removing F_0 and recomputing the frontier of the remainder yields F_1 , and iterating assigns every network to some layer F_k . The Pareto rank of a given network is the index k of its layer. Compared with binary frontier membership, this measure is better suited here because it is more robust to small perturbations of objective values, provides a graded summary of proximity to optimality.

Brain maps

We incorporated cortical brain maps provided through the neuro-maps Python library (112). Specifically, for the main analysis, we used four maps detailed below.

Functional connectivity gradient

The unimodal-transmodal gradient was derived from the first functional connectivity (FC) gradient introduced by Margulies and colleagues, which methodology is explained in details (60). Briefly, the FC1 gradient was derived from resting-state fMRI FC matrices and obtained using diffusion embedding. This principal gradient is anchored at one end by primary unimodal regions (visual, somatosensory/motor, and auditory) and at the other end by transmodal association regions, most prominently the default-mode network (DMN). Intermediate values correspond to heteromodal areas that integrate across sensory modalities (60).

SA gradient

The second gradient was the SA gradient from Sydnor *et al.* (56) and provides a unidimensional ordering of cortical regions from the primary sensorimotor cortex to the transmodal association cortex. This axis is constructed by rank-ordering parcels along multiple cortical features and averaging those ranks, yielding an archetypal gradient in which lower values index the sensorimotor-like cortex and higher values index the association-like cortex. Detailed methods and validation are provided in the source paper (56); however, briefly, Sydnor *et al.* derived the archetypal SA axis by aggregating parcel rankings from 10 heterogeneous data types (e.g., T1w/T2w myelin, FC principal gradient, evolutionary expansion, cortical thickness, aerobic glycolysis, receptor/transcriptomic proxies, etc.) computed on the 180 left-hemisphere parcels of the HCP multimodal atlas and then averaging these ranks to produce a single cortex-wide gradient that was used here.

FC homology gradient

The third gradient was the FC homology between humans and macaque monkeys (63). This map quantifies cross-species similarity of cortical functional organization between humans and macaques. Briefly, FC homology is built from a joint functional space via diffusion embedding of a concatenated similarity matrix. FC homology at each human vertex is computed as the maximum cosine similarity of the first set of gradients (top ~15) within a 12-mm searchlight around the surface matched location in macaque, yielding a vertex-wise index of functional homology. As reported in the source (63), FC homology is highest in unimodal systems (e.g., early visual cortex including V1/MT, auditory, and somatomotor) and lowest in the transmodal association cortex, with minima in DMN territories (e.g., angular gyrus, posterior medial cortex, and lateral temporal cortex), consistent with an evolutionary hierarchy from sensorimotor to association regions.

Neurosynth activation map of meta-analytic cognitive tasks

Last, to contextualize our findings with cognitive functions, we used the cortical map corresponding to the dominant principal component of Neurosynth terms that overlap with the Cognitive Atlas. This map was generated by Yarkoni *et al.*'s Neurosynth framework (61), which automatically aggregates neuroimaging studies to produce whole-brain association maps for psychological terms. Poldrack *et al.*'s Cognitive Atlas provides a curated ontology of cognitive constructs (62). The intersection of these resources yields 123 cognitive terms (e.g., attention, pain, working memory, and social cognition; full list in table S1), each associated with a meta-analytic cortical activation pattern. The first principal component of these maps captures the

dominant axis of variation, and the resulting cortical map assigns each region a score capturing its loading on this axis, situating them along an axis dominated by perceptual/motor tasks versus emotion and social (56, 113).

Null maps for statistical significance

Last, to assess the statistical significance of correlations between cortical surface maps while controlling for spatial autocorrelation and hemispheric symmetry, we used the spherical rotation (spin) test, also available in Neuromaps (112, 114). In brief, each cortical map was projected to a sphere, and then random 3D rotations were applied to the spherical coordinates to generate null maps that preserve spatial contiguity and smoothness but disrupt anatomical alignment. Rotations are applied identically within a hemisphere and with sign-constrained axes across hemispheres to preserve contralateral symmetry; vertices rotated off the cortex (e.g., into midline/corpus callosum gaps) are omitted or reassigned by nearest neighbor on the rotated sphere. The empirical correspondence statistic is compared to its permutation null formed by repeated spins. For each comparison, we performed $N = 1000$ independent spins. At each iteration k , a random rotation matrix R_k [uniform on $SO(3)$ with hemisphere constraints] was applied to X to obtain $X^{(k)}$, and the Pearson's correlation coefficient was recomputed. The (two-sided) spin-based P value was then estimated as

$$p_{\text{spin}} = \frac{1 + \sum_{k=1}^N \mathbb{1}[|r^{(k)}| \geq |r_{\text{obs}}|]}{N + 1}$$

yielding inference under the spatial permutation framework and controlling inflated false positives that arise from naive parametric tests on autocorrelated surfaces.

Supplementary Materials

This PDF file includes:

Figs. S1 to S12

Table S1

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available at: www.humanconnectome.org/study/hcp-young-adult/document/900-subjects-data-release, and the following open-source Python libraries were used: YANAT (Yet Another Network Analysis Toolkit; <https://kuffmode.github.io/YANAT/>) for the game-theoretic framework, Echoes (<https://fabridamicelli.github.io/echoes/>) for neuromorphic modeling, Netneurotools (<https://netneurotools.readthedocs.io/en/latest/>) for thresholding empirical networks, Brain Connectivity Toolbox (BCT; <https://github.com/aestrivex/bctpy>) for constructing degree-preserved null models, Networkx (<https://networkx.org/>) for constructing fully random networks, and Neuromaps (<https://netneurolab.github.io/neuromaps/>) for

comparing brain maps. A plug-and-play tutorial for the game-theoretic generative model, including features that were not explored here, is also available at: https://github.com/kuffmode/YANAT/blob/main/examples/generative_game_theoric.ipynb.

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